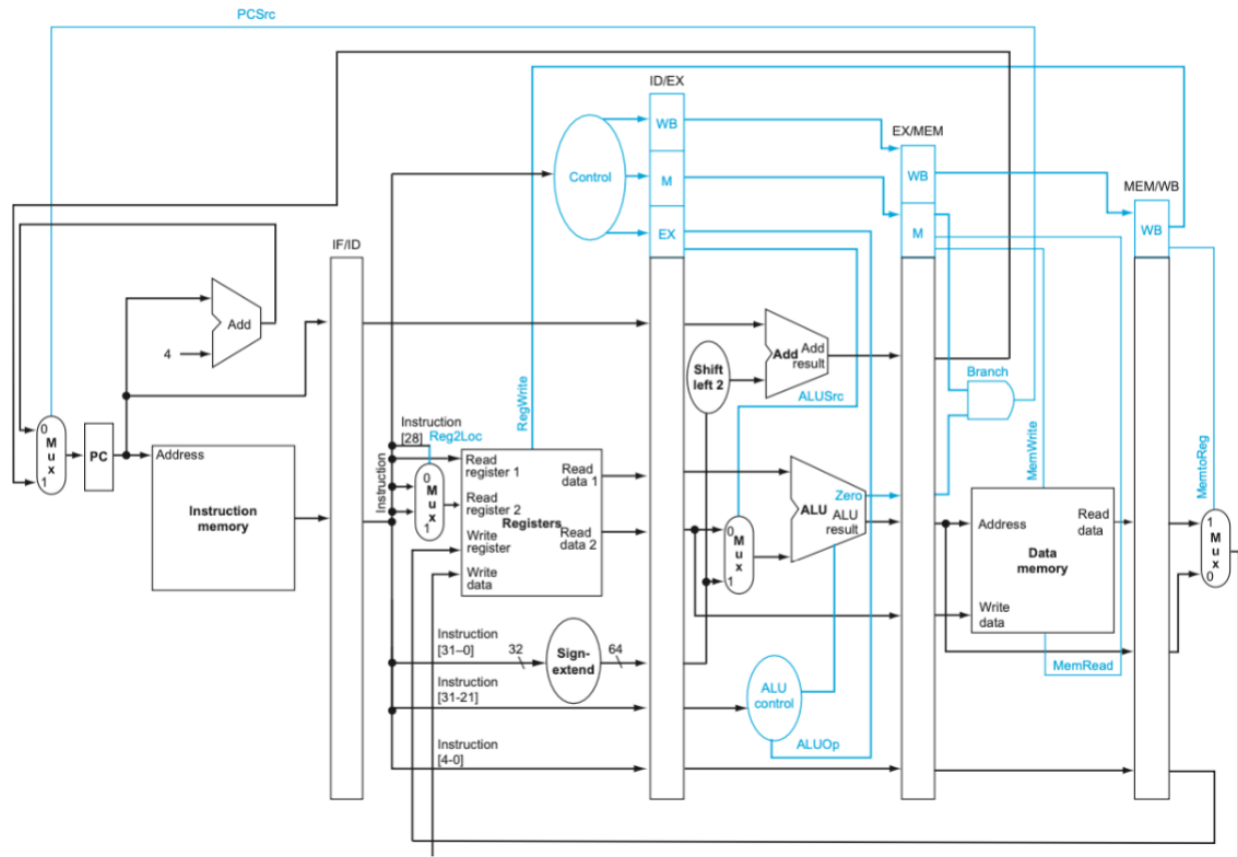


64-bit ARM 5-stage pipelined microprocessor with SystemVerilog



- 35 modules and 21 additional test benches written
- Handles these assembly instructions: ADDI, ADDS, SUBS, B, B.LT, BL, BR, CBZ, LDUR, STUR
- Constraints: Only control logic can have RTL, and any logics gates can have at most 4 inputs
- Notable challenges:
 - Designing and implementing additional architecture for B.LT, BL, and BR
 - Addressing pipeline hazards; coordinating between Forwarding module, Control module, Hazard module, pipeline registers, and all the multiplexors for flushing and/or forwarding data.
 - Debugging unexpected results by creating test benches and using ModelSim